

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 24

Design of IIR Filters from Analog Filters — Impulse Invariance

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of Impulse Invariance

1.1 Definition & Invariance Criterion

The **Impulse Invariance Method** designs a digital IIR filter by sampling the continuous-time impulse response $h_a(t)$ of an analog prototype filter at uniform sampling intervals T_s :

$$\mathbf{h}[n] = \mathbf{T}_s \mathbf{h}_a(n\mathbf{T}_s), \quad n = 0, 1, 2, \dots \quad (1)$$

The factor T_s scales the discrete response so that the digital filter's low-frequency gain matches the analog filter.

1.2 The Fundamental Pole Mapping Rule

Let the analog filter transfer function be expanded in partial fractions with distinct poles s_k :

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \iff h_a(t) = \sum_{k=1}^N A_k e^{s_k t} u(t) \quad (2)$$

Sampling at $t = nT_s$ and taking the Z -transform yields the **Master Operational Formula**:

$$\frac{\mathbf{A}_k}{s - \mathbf{s}_k} \xrightarrow{\text{Impulse Invariance}} \frac{\mathbf{T}_s \mathbf{A}_k}{1 - \mathbf{e}^{\mathbf{s}_k \mathbf{T}_s} \mathbf{z}^{-1}} \quad (3)$$

Every continuous pole at $s = s_k$ maps directly to a discrete pole at:

$$\mathbf{z}_k = \mathbf{e}^{\mathbf{s}_k \mathbf{T}_s} \quad (4)$$

2 Stability Mapping & The Aliasing Dilemma

2.1 Preservation of Stability

Let an analog pole be $s_k = \sigma_k + j\Omega_k$. The corresponding digital pole has magnitude:

$$|z_k| = |e^{(\sigma_k + j\Omega_k)T_s}| = e^{\sigma_k T_s} \quad (5)$$

- If $\sigma_k < 0$ (Left-Half s -Plane) $\implies |z_k| = e^{\sigma_k T_s} < 1$ (**Inside the Unit Circle**).
- **Theorem:** A stable analog filter is **always mapped into a strictly stable digital filter**.

2.2 Frequency Relationship & The Aliasing Problem

The normalized digital frequency ω is related to analog frequency Ω by:

$$\omega = \Omega T_s \quad (6)$$

The discrete frequency spectrum is an infinite periodic superposition of shifted analog spectra:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a \left(j \frac{\omega - 2\pi k}{T_s} \right) \quad (7)$$

Because the analog rational transfer function $H_a(j\Omega)$ cannot be strictly band-limited to $|\Omega| \leq \pi/T_s$, the high-frequency tails fold back into the baseband $[-\pi, \pi]$, creating ****aliasing distortion****.

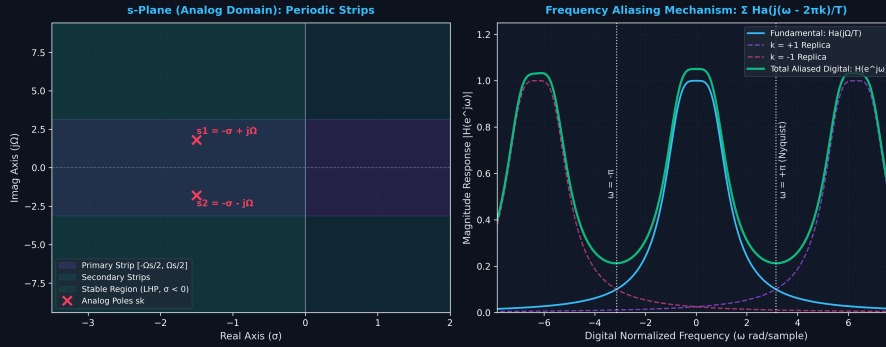


Figure 1: Left: Horizontal strips of width $2\pi/T_s$ in the s -plane mapping into the z -plane unit circle. Right: Aliasing caused by overlapping tails of the periodic spectrum copies.

- **Suitable Filters:** Lowpass and Bandpass filters whose analog stopband attenuates deeply (> 30 – 40 dB) at the Nyquist frequency $\Omega_N = \pi/T_s$.
- **Unsuitable Filters:** Highpass and Bandstop filters **CANNOT be designed** via impulse invariance because their high-frequency gains fold over into DC ($\omega = 0$), destroying stopband attenuation.

3 Step-by-Step Design Protocol

1. **Identify Sampling Interval T_s and Frequencies:** Convert digital specifications using $\Omega = \omega/T_s$.
2. **Factor the Analog Transfer Function:** Find all poles s_k of $H_a(s)$.
3. **Perform Partial Fraction Expansion:**

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k}, \quad \text{where } A_k = \lim_{s \rightarrow s_k} (s - s_k) H_a(s) \quad (8)$$

4. **Apply Pole Mapping Formula:** Replace each term by $\frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}}$.
5. **Combine Conjugate Terms:** Put complex conjugate pole pairs over a common quadratic denominator to yield real biquad filter coefficients.

4 Standard Solved Exam Example

Problem: An analog second-order Butterworth lowpass filter with cutoff $\Omega_c = 1$ rad/s is given by:

$$H_a(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad (9)$$

Design the corresponding digital IIR filter using the Impulse Invariance method with $T_s = 1$ second.

Solution:

Step 1: Find the Analog Poles

$$s^2 + \sqrt{2}s + 1 = 0 \implies s_{1,2} = -\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}} \approx -0.7071 \pm j0.7071 \quad (10)$$

Step 2: Partial Fraction Expansion

$$H_a(s) = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} \quad (11)$$

Evaluating the residues:

$$A_1 = \frac{1}{s_1 - s_2} = \frac{1}{j2(0.7071)} = \frac{1}{j\sqrt{2}} = -j\frac{1}{\sqrt{2}} \approx -j0.7071 \quad (12)$$

and $A_2 = A_1^* = +j\frac{1}{\sqrt{2}} \approx +j0.7071$.

Step 3: Map Poles to the Z-Domain ($T_s = 1$ s)

$$z_1 = e^{s_1 T_s} = e^{-0.7071 + j0.7071} = e^{-0.7071} \angle (0.7071 \text{ rad}) \quad (13)$$

Converting $0.7071 \text{ rad} = 0.7071 \times \frac{180^\circ}{\pi} \approx 40.51^\circ$:

$$e^{-0.7071} \approx 0.4931, \quad \cos(40.51^\circ) \approx 0.7602, \quad \sin(40.51^\circ) \approx 0.6496 \quad (14)$$

$$z_1 = 0.4931(0.7602 + j0.6496) \approx \mathbf{0.3748 + j0.3203}, \quad z_2 = z_1^* = \mathbf{0.3748 - j0.3203} \quad (15)$$

Step 4: Combine into Real Second-Order Biquad

$$\begin{aligned} H(z) &= \frac{T_s A_1}{1 - z_1 z^{-1}} + \frac{T_s A_2}{1 - z_2 z^{-1}} = \frac{-j\frac{1}{\sqrt{2}}(1 - z_1^* z^{-1}) + j\frac{1}{\sqrt{2}}(1 - z_1 z^{-1})}{(1 - z_1 z^{-1})(1 - z_1^* z^{-1})} \\ &= \frac{j\frac{1}{\sqrt{2}}(z_1^* - z_1)z^{-1}}{1 - 2\text{Re}(z_1)z^{-1} + |z_1|^2 z^{-2}} \end{aligned} \quad (16)$$

Evaluating the numerator and denominator parameters:

- $z_1^* - z_1 = -j2\text{Im}(z_1) = -j2(0.3203) = -j0.6406$
- Numerator coefficient: $j\frac{1}{\sqrt{2}}(-j0.6406) = \frac{0.6406}{\sqrt{2}} \approx \mathbf{0.4530}$
- Denominator z^{-1} coefficient: $2\text{Re}(z_1) = 2(0.3748) = \mathbf{0.7496}$
- Denominator z^{-2} coefficient: $|z_1|^2 = (0.4931)^2 \approx \mathbf{0.2431}$

The final digital filter transfer function is:

$$\mathbf{H(z) = \frac{0.4530z^{-1}}{1 - 0.7496z^{-1} + 0.2431z^{-2}}} \quad (17)$$

Stability verification: Poles are at $r = |z_{1,2}| = 0.4931 < 1$, ensuring a strictly stable digital filter.

5 Essential Exam Checklist & Traps

- **Partial Fraction Rule:** Always expand $H_a(s)$ into partial fractions first. Never substitute $s = \frac{1}{T} \ln(z)$ algebraically into $H_a(s)$.
- **The Scaling Factor T_s :** Always include T_s in the numerator: $\frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}}$.
- **Zero Locations:** Remember that **zeros do not map** via $z = e^{s T_s}$. Only poles obey this mapping.
- **Highpass Prohibition:** Impulse Invariance cannot be used to design Highpass filters due to infinite spectral foldover (aliasing) at DC.